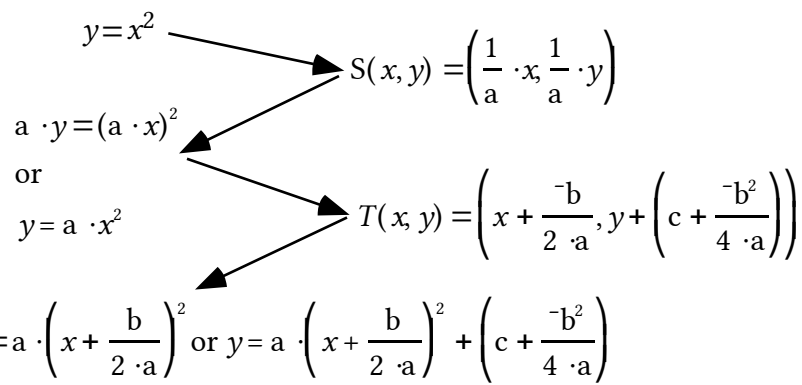
$y = x^2$
$S(x,y) = \left(\frac{1}{a} \cdot x, \frac{1}{a} \cdot y\right)$
$a \cdot y = (a \cdot x)^2$
or
$y = a \cdot x^2$
$T(x,y) = \left(x + \frac{-b}{2 \cdot a}, y + \left(c + \frac{-b^2}{4 \cdot a}\right)\right)$
$y + -\left(c + \frac{-b^2}{4 \cdot a}\right) = a \cdot \left(x + \frac{b}{2 \cdot a}\right)^2$ or $y = a \cdot \left(x + \frac{b}{2 \cdot a}\right)^2 + \left(c + \frac{-b^2}{4 \cdot a}\right)$